

Systems integration and the environment

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Governments and businesses alike are increasingly taking account of environmental concerns in their own operations, with an overall objective of moving towards a sustainable society, i.e. one that meets the needs of today without compromising the well-being of future generations. Telecommunications has not been viewed as an activity that causes severe environmental problems and has, therefore, been omitted from the debate at reaching solutions to current problems. The collaborative project described in this paper, in which BT played a leading role, undertook to produce a network operators' charter which would commit telecommunications' providers to certain minimum levels of environmental behaviour. To meet these levels, internal activities have been assessed and best practices developed for adoption. The commercial and societal benefits in using telecommunications services were also modelled to provide an estimate of the advantages to be gained.

1. Introduction

At the Earth Summit in Rio de Janeiro in 1992, heads of government from all over the globe debated the issues and committed their countries to action towards the goal of sustainable development. Indeed, many states subsequently published national strategies for how they intended to achieve that goal. What has yet to be developed, however, is the global vision that is needed to draw all the individual strands into a comprehensive and workable whole. As the Chair of the Board of the European Environmental Agency (EAA) has commented [1]: 'The Rio aspiration for a major compact between countries of North and South to build a partnership for sustainable development remains elusive'.

A number of official European Union (EU) and national government documents have already identified areas where the telecommunications industry can be associated with both positive and negative environmental impacts. Of particular note is the recent EU green paper on the liberalisation of telecommunications infrastructure and cable television networks which includes a clause that requires, as a license condition, an in-depth assessment of the negative impacts of telecommunications infrastructure projects. As well as minimising potential threats, however, the goal of sustainable development can be brought much closer through the use of telecommunications services. At the same time, there are tremendous potential benefits to be gained by service providers.

A collaborative project, under the auspices of EURESCOM — the European Institute for research and strategic studies in telecommunications — started in September 1995 to formulate a common European position for the public network operators (PNOs) on environmental

issues and sustainable development. The objectives of the project were:

- the production of an environmental policy paper which individual PNOs would sign, committing them to make environmental improvements within their own activities, combined with an active contribution to sustainable development,
- the identification of common areas of improvement, with associated guidelines for action, environmental performance indicators and reporting mechanisms,
- the identification and systematic analysis of commercial opportunities arising from sustainable development.

The project involved a total of 12 partners: Belgacom, BT, CNET, CSELT, Portugal Telecom, Telenor, Telefonica, Telia, OTE (Greece), Deutsche Telekom, Swiss Telecom and KPN (the Netherlands). The project itself lasted for 18 months. BT was represented on the project by Dr Chris Tuppen, leader for the overall project, and three people from Technical Facilities (TF), a unit within Systems Integration at BT Laboratories (BTL). The placing of the environmental unit within Systems Integration was no accident. The issue of environmental impacts will become increasingly important for telecommunications providers and the Systems Integration unit is directly involved with most of the major new systems and applications to be introduced into BT.

The paper is structured to follow the three separate sections of the project itself — the environmental charter, internal PNO activities (including procurement, energy and waste management and training), with the final section focusing on the societal and commercial benefits to be gained from the deployment of telecommunications services and applications. As an overall introduction to the project, however, there is an initial section on sustainable development, the guiding principle for all the work that was undertaken.

2. Sustainable development

In 1972, the Club of Rome published 'Limits of Growth', the now famous first report on its project on the Predicament of Mankind [2]. Twenty years later, a sequel, entitled 'Beyond the Limits', found the three conclusions drawn in the first report to be '...still valid, but they need to be strengthened'. In summary, these conclusions are as follows.

- Human use of many essential resources and the generation of pollutants have already surpassed rates that are physically sustainable. Without significant reductions in material and energy flows, there will be an uncontrolled decline in per capita food output, energy use and industrial production.
- The decline is not inevitable. To avoid it, two changes are necessary. The first is a comprehensive revision of policies and practices that perpetuate growth in material consumption and population. The second is a rapid, drastic increase in the efficiency with which materials and energy are used.
- A sustainable society is technically and economically possible. It is much more desirable than a society that tries to solve its problems by constant expansion. The transition to a sustainable society requires a careful balance between long-term and short-term goals and an emphasis on sufficiency, equity, and the quality of life rather than on quantity of output. It requires more than productivity and more than technology; it also requires maturity, compassion and wisdom [3].

Such concerns have led to the search for an alternative vision that would help society resolve the increasingly apparent contradiction between economy and environmental protection. As a consequence, the concept of sustainable development was adopted, with its most enduring definition provided by the 1987 Brundtland Report.

'[Sustainable development] meets the needs of the present without compromising the ability of future generations to meet their own needs.' [4]

Despite the fact that many nations have published their own strategies for securing sustainable development, the task of achieving it has, so far, defeated everyone involved with it. Indeed, how can one protect the environment and improve the quality of life for all people when the very economic system that caused the problems in the first place remains more or less in operation?

This is a tremendous challenge, requiring a collective effort on a global scale that is quite unprecedented. Fortunately, there are new ways of looking at the problem that banish the negativity with which the challenge is usually reported. For example, the World Bank has developed a new way of looking at the world's economy. This approach makes it easier to foresee the convergence between increasing human welfare and protection of the environment [5]. Their increasingly popular method of analysing an economy recognises at least four categories of capital, as shown in Table 1.

Table 1 The World Bank's four categories of capital.

Natural capital	'Mined' material such as fossil fuels and minerals; biological diversity and productivity; 'services' such as absorption of pollution.
Human-made capital	Machines, factories, buildings, infrastructure and money.
Human capital	Investments in education, health and nutrition.
Social capital	The institutions and cultural bases for a society to function.

The Bank avers that sustainable development will mean an economy that is much less dependent on natural capital, and uses human-made capital very differently in order to achieve the enormous potential for growth in both human and social capital. Since an economic solution seems possible, is it also technologically feasible?

The view that is receiving greater recognition now is that there are two technologies with a significant potential for making a telling contribution towards sustainable development. These technologies are energy generation from renewable resources, and telecommunications. The EURESCOM project sought to investigate the opportunities for telecommunications to make a difference.

3. The Environmental Charter

Among PNOs, BT has already set standards for reporting on environmental activities. The Environmental Report produced by BT's corporate environmental issues group has already won a number of awards. In this respect, BT is already further advanced than other European PNOs. It is important, however, not to stop at that stage.

A number of other organisations, e.g. banks, have used the medium of an environmental charter to demonstrate

their commitment to improvement. Such a move, however, needs to be followed through. Actual improvements need to be demonstrated, otherwise the document becomes a dead letter and exists merely for show.

The Charter provides a telecommunications operator with a mechanism for overall co-ordination of environmental activities and a strategy to guide it towards management structures that incorporate sustainable development concepts. The decision to sign the Charter, however, is not something to be taken light-heartedly. Signing the document means a full understanding of the implications it may have on company decisions at various levels. It means a commitment not only to act in words but also in deeds. Table 2 shows the terms of the European Telecommunications Network Operators (ETNO) Charter.

One of the objectives for signing the Charter is to gain social, technical and commercial know-how about environmental opportunities and to turn them into an asset rather than a burden. The most significant benefit, however, from the adoption and promotion of the Charter is to give the telecommunications industry an underlying credibility in terms of promoting the use of telecommunications services as a highly effective means of achieving some of

the goals connected with sustainable development. This should offer significant market potential for telecommunications operators. However, their promotion would be extremely dangerous without the underlying commitment of PNOs to minimise the impact of their own operations on the environment.

From an individual PNO perspective, the benefits from embarking on a programme of environmental improvement range from meeting customer expectations, through financial gain from energy and waste reduction, to minimisation of future liabilities such as contaminated land.

The six goals of the Charter will only be reached through corporate awareness at the highest level, with care for the environment being defined as a priority in all management decisions, present and future. Education and training of people will be essential, while encouraging our suppliers to adopt a similar approach will be a true challenge. A total of 21 PNOs are current signatories. The Environmental Charter itself was signed by Dr Alan Rudge, on behalf of BT, in November 1996.

An ETNO Working Group, led by Dr Chris Tuppen, is currently seeking to define the criteria that will be used to

Table 2 The ETNO Charter.

Environmental Charter of European Telecommunications Network Operators	
Our vision	
Sustainable development is a strategic global environmental goal. It describes development that takes into consideration the need to conserve both the natural environment and the world's scarce non-renewable resources for future generations. It is our belief that we can play an important part in making this happen.	
This Charter describes our commitment to sustainable development through:	
• the provision of products and services that provide significant environmental benefits, • a determination to manage our own operations in a way that minimises negative environmental impacts.	
Our Approach	
We recognise that the universal presence of telecommunications in today's society places on us a social obligation to be good corporate citizens. A responsible attitude to environmental issues is an important part of meeting that obligation.	
As a collective group of companies, our combined turnover represents a significant proportion of European trade and this puts us in a unique position to make a real difference.	
Our Commitment	
Whether together, or individually, we are committed to continuous improvement through action in the following areas:	
<i>Awareness</i>	
We shall aim to ensure recognition and acknowledgement of all relevant environmental impacts, including the positive and negative impacts of our products and services. In particular, we shall build the environment into our training programmes and company communications programmes.	
<i>Regulatory Compliance</i>	
We shall strive to achieve full compliance with all relevant environmental legal requirements, and to exceed these requirements where appropriate.	
<i>Research and Development</i>	
We shall support research and development into the contribution new telecommunications services can make to sustainable development.	
<i>Procurement</i>	
We shall build environmental considerations into our procurement processes. Special attention will be paid to energy consumption, waste management, process and product requirements and the use of hazardous materials.	
<i>Providing Information</i>	
We shall provide relevant data and information about our environmental performance to employees, customers, shareholders and governments.	
<i>Environmental Management Systems</i>	
We shall implement environmental management systems that support the development of appropriate and well-structured environmental protection.	

report on the progress being made by individual PNOs. From 1998 onwards, this Working Group will prepare an annual report based on information supplied by the signatories. The report will demonstrate how well the Charter is being followed and what achievements have been made.

4. Identification of common areas of environmental improvement

This part of the project was split into a number of separate areas, covering procurement, waste management and minimisation, energy, and education.

4.1 Guidelines for environmentally responsible procurement

All production of products and services, however it may be achieved, uses energy and materials that impact upon the environment. Many of these effects are negative in relation to the ideal conditions envisaged for sustainable economic and social prosperity. Any organisation wishing to support the principles of sustainable development must work to reduce the negative impact on the environment of its own activities.

PNOs are large organisations that purchase, consume and deploy vast volumes of materials in one form or another. They therefore have a responsibility to consider the environmental impact generated throughout the whole life cycle of the materials used in their products, i.e. from initial production, through manufacture into the product or services to final disposal. The procurement function is the main interface and enabler of the passage of materials into a PNO, and this interface may be used to channel environmentally responsible specifications to suppliers and to verify that the products and services have been produced in an environmentally responsible way. One way of achieving this is for PNOs to generate and apply common environmental guidelines through the procurement process. This would have the advantage of setting out a common set of requirements to suppliers, detailing certain minimum criteria and of using the massive purchasing power of the European PNOs to secure commercial benefits. Further, environmental criteria could be used as part of the supplier selection process. BT already has a process in place to achieve this, but the outcome of the EURESCOM project is being used to enhance this system.

The process produced within the project consisted of five distinct steps that would allow a PNO with no previous knowledge of environmental procurement guidelines (EPGs) to develop a coherent and effective approach to environmentally responsible procurement. These steps are:

- a definition of the purpose and fit of EPGs within the organisation (this step establishes a rationale and focus for the development of EPGs),
- a description of the elements of ownership required for successful management of EPGs, including those elements that are critical for policy development,
- a description of the critical factors that need to be addressed within the PNO before applying EPGs to suppliers,
- a description of the critical factors which need to be addressed in the operational application of EPGs to suppliers,
- an explanation of the way in which a self-assessment audit of infrastructure and process development within the organisation relating to environmentally responsible procurement can be conducted — this audit process is designed to highlight areas where further development is required and to assist in the construction of an action plan.

It should be understood, however, that EPGs are unlikely to be effective unless they have been constructed to operate within the overall management framework of the organisation. EPGs must be embedded into the procurement process, but must also be generated according to the framework laid down by the organisation's quality management system (QMS) and procurement management system (PMS). If the organisation additionally operates an environmental management system, EPGs will need to be congruent with them also. EPGs should not be derived directly from the environmental management system (EMS) because they deal with specific hardware and materials issues. However, the EPGs must be compatible with the EMS and their application should be traceable through both the EMS and QMS.

Having established the place of EPGs in the system, the next stage involves establishing the ownership of EPGs within the company and the top-level policy statements that will apply. These policy statements will be used to derive guidelines against which performance can be assessed. It is recommended that EPGs are owned and championed at the same level within the PNO as the PMS. The top-level policy statements that should then be produced should cover the following areas:

- reduction of the environmental impact connected with all procurement activities,
- alignment of procurement standards and guidelines within an overall EMS,
- embedding of environmental procurement standards and guidelines into the supply chain,

- adoption of cradle-to-grave responsibility for procured products and services,
- minimisation of the use of materials, supplies and energy in procured products and services,
- encouragement of the use of renewable and recycled materials and components where it is technically and economically feasible, and safe, to do so,
- using supplier selection and management as a means of minimising the waste produced in all parts of the business as a result of the procurement of products and services,
- aiming for continuous improvement.

A self-assessment template for use by all procurement organisations has been produced comprising a total of 48 questions and 4 performance categories. The assessment category will then provide an indication of the action that a PNO should consider making in order to achieve improvement.

4.2 *Eco-design model*

PNOs are major users of hardware in their own networks and major suppliers of equipment to their customers. In terms of asset management and obligations for end-of-life activities, this represents a tremendous responsibility and cost. By considering the end-of-life issues when a piece of equipment is designed that responsibility can be controlled better and the potential costs, e.g. for disposal, can be significantly reduced. Indeed, by designing in the best attributes, opportunities for generating revenue from hardware after it has reached the end of its life within the PNO can be produced.

The most likely end-of-life paths that can exist for hardware are:

- scrapping — this will probably be to landfill or incineration, with no recovery of materials and little likelihood of revenue from the product itself (there may be some energy recovery as a result of incineration, but it is unlikely that the benefits from that energy will accrue to the PNO),
- recycling — this will require the destruction of the original purpose of the product, some revenue possibly generating from the materials and/or components that are obtained, but it is quite likely that there will be an overall cost to the PNO in performing this operation,
- re-manufacture — this involves the restoration of the product to its original or near original state, the process involving re-work and the production of replacement items (with a requirement for additional use of materials), but it is likely that increased revenues can be gained over and above those that might accrue from recycling,

- re-sale — this involves selling-on the product to a new owner, the process usually involving the minimum possible re-work and probably yielding much higher revenues than the options listed above,
- upgrading — this involves adding functionality to the product, usually without a change of ownership, extending the life of the product in the most efficient way.

If the life cycle of a product is planned, it should be possible to include the best set of design criteria for the chosen end-of-life option when that product is originally conceived. This would mean that only those design options that suit the chosen end-of-life path would be included. For instance, if a product were to be scrapped there would be no need to include features that would promote re-sale in the design. In this way, the actual costs of the design could also be controlled.

The outcome of the work in this section of the project was the production of a matrix of design criteria that could then be applied to optimise the design of a product for its chosen end-of-life. The matrix will only provide optimum benefit, however, when a PNO chooses the end-of-life option. Also, since all products will ultimately reach a point where their utility can no longer be extended, the matrix uses recycling of materials/components as the default.

Using the matrix involves three simple steps. The first step is completion of Table 3 with its rules and possible end-of-life options. Where a guideline is appropriate to a particular end-of-life option, an asterisk appears. Some of the guidelines will be appropriate to only one end-of-life option while others may be suitable for all five. Having chosen the appropriate end-of-life option for the product under consideration, that product is then evaluated against each of the attributes. Where the design has the particular attribute, a 'Y' is inserted. Where the attribute is not present, an 'N' appears.

The second step requires that all the attributes for the chosen end-of-life option where the 'N' appears are then reviewed so that the design can be amended to include as many of these options as possible. The final step involves a final review of the design to determine where further improvements may be made. An example of the use of the matrix appears in Table 3.

In the example shown, where the default option of recycling has been used, a reconsideration of the responses for the first design allowed a number of changes to be made so that only one of the rules ended up being broken. The rules themselves have been produced with customer premises equipment mainly in mind and may need further amendment to be truly generic.

Table 3 End of life options matrix.

Generic design rules	End-of-life options					Designs	
	A	B	C	D	E	1	2
1. Availability of spares and consumables			*	*	*		
2. Cleanable to first market standards			*		*		
3. Design for second markets			*	*			
4. Durable high value sub-assemblies				*	*		
5. Durable skeleton and core technology				*	*		
6. Easy access to replaceable parts			*	*	*		
7. Easy separation of contaminated materials	*	*			*	Y	Y
8. Easy to change logo				*			
9. Easy disassembly into constituent parts	*	*			*	Y	Y
10. Avoid the use of glue and adhesive tape		*		*	*	N	Y
11. Design electrical circuits in a modular fashion		*		*	*	Y	Y
12. Design keypad for disassembly		*			*	Y	Y
13. Field maintainable			*	*			
14. Fits into standard packaging		*	*		*	Y	Y
15. Materials identified and marked	*	*	*	*	*	Y	Y
16. Maximise density	*						
17. Maximise durability (look and function of external and moving parts)				*	*		
18. Minimal value	*			*	*		
19. Minimise controlled and potentially controlled materials	*	*	*	*	*	Y	Y
20. Minimize material mixtures	*	*				N	Y
21. Minimise materials contamination	*	*				N	Y
22. Minimise mass and volume (with reference to collections costs)	*	*				Y	Y
23. Modular design				*	*		
24. Parts identified with function and number of lives		*		*	*	N	N
25. Use recycled materials wherever possible		*				N	Y
Application design rules							
26. Customer-friendly upgrade				*			
27. Instructions/markings readable in second market				*			
28. Software feature options			*	*	*		
29. Standardised interfaces			*	*			
30. Built-in compatible functions for second market			*				
31. Testable sub-assemblies			*	*	*		
Design 1 RULES MET RULES BROKEN						8 5	
Design 2 RULES MET RULES BROKEN							12 1

Key: End-of-life options A Scrap B Recycle C Resale D Upgrade E Re-manufacture

4.3 Waste management and minimisation

Waste management is usually understood as the identification of the best methods of disposal for a particular product. In more environmentally aware organisations, however, it comprehends other factors as well, including an understanding of the materials involved and their impact on the environment. Such an understanding is then used to highlight those materials that are either hazardous or could have a harmful effect on the environment, and cause them to be replaced by more environmentally friendly alternatives.

Waste minimisation, on the other hand, involves an overall reduction of the waste that is produced by/within an organisation. Because PNOs already have a considerable

portfolio of products in the network, some of which will contain materials that are now classified as hazardous (but which were not when they were purchased), there is a potentially serious waste problem with these products. At the same time, PNOs are still purchasing equipment. In terms of waste minimisation, therefore, there are two positions to be considered:

- for products that have already been procured, the disposal of every item must be considered carefully;

— the hierarchy of end-of-life options needs to be considered, with disposal to landfill as the final option,

— where landfilling is the only available option, proper routes need to be identified, depending on the materials involved, so that the costs to both the PNO and the environment are minimised;

- products that will be procured in the future should be specified and designed with the desired end-of-life option being taken into consideration (see above), the future costs of waste management thus being minimised, for both the PNO and the environment.

Both of the above positions aim to minimise the amount of virgin material that will be used to manufacture new products. If followed they will have a significant impact on the levels of contamination by hazardous materials and reduce the net amount of material that is ultimately sent to landfill. This would result in significant savings to the PNO in the long run. In order to achieve all these outcomes, however, waste minimisation must be built into the PNO's management system.

In order to put an effective waste minimisation system in place, the PNO's corporate environmental policy statement should state its commitment to the prevention of pollution caused by uncontrolled disposal and a reduction of the amount of waste generated through its own activities. In terms of the operation of such a system, however, the appropriate information needs to be assembled so that the system can either be built or improvements can be made. In assembling that information, a whole series of questions should be addressed. Indeed, the project highlighted 17 specific areas that needed to be fully considered.

The approach used is congruent with the requirements of ISO 14001 [6]. It does not claim to be exhaustive but, when completed, will provide the necessary information for a PNO to establish the current level of control of waste and focus on those areas where improvements are necessary.

4.4 Energy management

Energy is a very large and visible cost for a PNO. At the same time, depending on how that energy is generated, there is also an environmental cost to be paid. This suggests that there are two distinct areas where a PNO's energy management policy can rest — on the control and minimisation of the energy that is used *per se* and on the employment of less polluting means of providing the energy that is used. The work within the project covered both these aspects.

In terms of the control and minimisation of energy use, a set of benchmarks were produced so that a PNO could know its own position, and be able to compare it with other PNOs, regardless of the state of development of the network, size, climate of the country, etc. These indicators appear in Table 4.

Table 4 Energy management indicators.

Indicator	Consumption Factor	Comments
E_T T	This is the energy consumption relating to the whole of a PNO's operation.	This indicator only applies to the consumption of a single PNO, i.e. it does not take into account the energy required in other PNOs for international traffic.
E_S L	This is the energy used in switching and can be related to the whole of a PNO's operation or any segment, e.g. a zone or a building.	It includes voice and data traffic using the switched network. It does not include, however, the mobile network.
E_C $C \times L$	This covers the energy used in cooling equipment. It can be used to cover the whole of a PNO's operation or a segment.	The factor C is the sum over the time period chosen of the average temperature difference, i.e. outdoor air temperature minus indoor air temperature.
E_M $L \times U$	This covers the energy used by equipment in the mobile network.	The factor L represents the coverage of the system, while U covers usage.

Key:

E_T is the total annual energy consumption.

T is the number of telephone lines.

E_S is the energy consumed in switching.

L is the number of customer lines.

E_C is the energy used in cooling.

C is the cooling day degree.

E_M is the energy consumption in the mobile operation.

U is the number of charged units.

Once a PNO has established its actual consumption it can then choose measures to make reductions, as appropriate. However, the energy that a PNO uses has probably been produced from fossil fuels or by other means that are environmentally unfriendly. The area where considerable improvements can be made in the energy field is through the use of so-called renewable energy.

The main source of renewable energy is the sun. Wind and water are additional sources. These all have the advantages that they can be adapted to local resources and needs, do not entail greenhouse effects, do not produce major wastes in themselves, have no major risks associated with them and generate no significant local pollution.

The renewable energy sources themselves can be categorised as follows:

Table 5 Suitability of clean technologies.

Renewable energy source	Energy provided	Power range	General use	Suitable for PNOs
Biomass	electricity heat	kW	heating	Yes
Biofuel	mechanical	kW	vehicle power	Yes
Wind energy	electricity	W to MW	power station, stand-alone power supply	No Yes
Photovoltaic	electricity	W to MW	power station, stand-alone power supply	No Yes
Solar thermal	heat electricity	W to kW MW	heating power station	Yes No
Hydro-electricity	electricity	W to MW	power station, stand-alone power supply	No Yes
Sea/tidal energy	electricity	MW	power station	No
Geothermal	electricity heat	kW to MW	power station	No

- biomass — the energy is generated from vegetation, either through processing or combustion,
- biofuels — the energy is generated from wood or plants such as sunflower to produce alcohol, etc, as a substitute for or additive in alcohol,
- wind energy — the energy source is the wind, turbines being used to convert mechanical energy into electricity,
- photovoltaic — the energy source is the sun, with sunlight being directly converted into electricity using mono, multi-crystalline or amorphous silicon,
- solar thermal — again, the energy source is the sun, with fluids contained in plan collectors being heated to produce steam energy,
- hydroelectricity — the mechanical power from water flow is converted directly into electricity,
- tidal/wave power — this employs the same principle as hydroelectricity, i.e. water flow, but is generally less advanced as a technology,
- geothermics — this involves using the earth's internal energy and is usually used near volcanic areas.

Although a wide range of possible energy sources exist, they are not necessarily all immediately accessible by PNOs. Indeed, as well as availability, the development of the particular technology connected with one or other of the energy sources may not be sufficiently well developed to provide an adequate alternative. Further, the political system may require that energy be produced in a particular fashion. Finally, the cost of using some of these systems is still not seen as being commercially viable.

Because of these factors, the potential for using clean technologies has not been exploited as much as it could. There is scope, therefore, for significant advances in this field.

Assuming that clean technologies were available to PNOs, however, it would be useful to know how effective they might be in supplying power requirements. Such an analysis appears in Table 5.

Of course, even where renewable energy may not be directly of use within a PNO, the fact that the PNO is taking its energy from a generating company that is using renewable energy as its source of power also has environmental advantages. As stated earlier, the view that is receiving greater recognition now is that energy generation from renewable resources has significant potential for making a telling contribution towards sustainable development.

4.5 Training

It is a truism that, whatever the effort that is put into processes it is the way in which they are operated that yields the eventual benefits. Environmental activities are no different from other processes in obeying this truism. Thus, if there is no commitment from the highest levels down through the company, the PNO is wasting its time in having environmental processes at all. Once that commitment is present, and the processes are in place, the workforce has to be motivated to understand not only what is meant in working with the processes but also what is the overall point in their existing at all. This requires education and training. The education and training that will be necessary for each individual in the workforce will, however, depend on their precise function. For those with particular environmental responsibilities, the amount of training required will be greater. There is, however, a certain minimum level that will need to be provided to every person within the company.

The project team investigated the training requirements to ensure a certain minimum level of expertise provision.

The recommendations were that there should be a whole range of components that comprised any training package. These components would depend on the level of responsibility within the company and the particular activities that the individual employee needed to undertake. Essentially, however, there were three levels of expertise provision:

- for those with environmental management responsibilities, the training would need to be the most comprehensive,
- for managers within the company, the training should cover not only their own legal obligations and those of the people working for them but, perhaps more importantly, they should be energised to ensure that their workforces understand the purpose of the requirements that have been placed on them and that they want to achieve these requirements,
- for the workforce itself, i.e. the people who have to employ the processes that have been provided for them — without their buy-in, environmental standards will remain low, and training for these people, therefore, must be directly related to the tasks that they have to achieve.

5. Commercial and societal benefits

As well as providing the potential for increased environmental benefits, telecommunications services can also have significant commercial benefits for the service providers. Therefore one team within the project focused on:

- the identification of those telecommunications services/applications that could result in significantly reduced environmental impact,
- the objective and quantifiable assessment of the societal benefits and drawbacks of these services/applications,
- the evaluation of the commercial opportunities afforded by sustainable development for PNOs.

Initially, the team identified over twenty individual services/applications that PNOs already, or could in the future, offer. Rather than consider each one individually, however, they were grouped together into a 'basket' of services aimed at a particular 'opportunity area'. This not only made subsequent calculations easier but also suggested a possible opportunity for marketing. The opportunity areas are defined below.

- Opportunity area 1 — letter post

This area includes the replacement of ordinary mail, invoices, bookings for travel/entertainment, literature/journals/newspapers, and advertising materials, etc. It

does not cover parcel post. In addition to deliveries by the national postal administration, this category includes non-parcel deliveries by other organisations, e.g. couriers, newspaper delivery persons, telephone directories and junk mail.

- Opportunity area 2 — travel options

This area covers business travel, commuting to work and other occupational uses of vehicles, e.g. the vehicle fleets of business organisations, public administrations, etc.

- Opportunity area 3 — entertainment/information transfer

This area covers the non-occupational options for minimising travel. It includes 'on-line' ordering of goods, with the consequent replacement of commercial activities (such as video stores), physical goods, e.g. CDs, audio/video cassettes, catalogues, etc, and the consequent travel requirements. It also includes 'on-demand' services such as tele-education and information transfer that will satisfy a client's needs and obviate the requirement for travel. Further, there are potential opportunities for new services such as environmental management systems and virtual reality. It should be understood that these services might not necessarily be available in an individual's home. Some travel may, therefore, be necessary.

- Opportunity area 4 — consumer services

This area covers those services directed towards the domestic customer. It includes 'on-line' ordering of goods that will subsequently require physical delivery, 'on-line' interactive access to personal data information, e.g. a bank account, 'on-line' booking of services, and the remote metering of domestic appliances and utility meters.

- Opportunity area 5 — telematics

This area includes the management of industrial processes, manufacturing activities and buildings, and will also cover disaster mitigation/management. This area focuses mainly on the rational use of resources through *in situ* control and monitoring. The major effect is in saving energy and materials.

Quantification of societal benefits from the substitution of telecommunications services into the identified opportunity areas was not deemed practical within the project. However, the general opinion was that the potential savings far outweighed the costs. Indeed, the scope for the services to make a difference in the identified opportunity areas may be imagined through considering just some of the published figures relating to current activities.

Data from the Universal Postal Union [7] shows that over 86×10^9 letter post items are handled each year in Europe. If one assumes a typical weight of an item to be 20 grams, this means that the amount of paper involved is greater than 1.7 million tonnes. Even the relatively small level of replacement of 25% of letter post by the identified services would mean that half a million tonnes of paper would not need to be produced. If newspapers and other documents printed on newsprint were 'published' via the World Wide Web, there could be a further estimated saving in paper production of around 10 million tonnes per annum.

Of course, paper is only one example where materials may be substituted by electronic media. The 'on-demand' services represent other obvious areas, where access to music, films and other forms of entertainment can be provided to individuals, thereby removing the need for personal ownership and, as a consequence, the deployment of materials.

Another obvious area where telecommunications can be deployed immediately and provide environmental benefits is in the relief of transport-derived pollution. There are a number of related issues that need to be understood when considering pollution caused by vehicles. The simplest concept is that relating to the chemicals produced during the combustion of fuel. These chemicals have been shown to be responsible for respiratory illnesses. They also damage buildings, forests and lakes, and contribute to the mass of greenhouse gases that are enabling global warming. All these elements have costs associated with them. Where efforts are made to clean up the effects of the polluting chemicals a further bill has to be paid. On top of this, of course, is the energy used and pollution caused in extracting, refining and distributing fuel in the first place.

Figures from the European Environmental Agency (EEA) [8] show that approximately 40% (7.2×10^9 tonnes) of the volatile organic compounds emitted in Europe are as a direct result of the extraction, distribution and use of fossil fuels. Further, these same fossil fuels, when used as transport fuels, contribute one-sixth (612×10^9 tonnes) of the total level of CO_2 — a potent greenhouse gas — and one-half (6.8×10^6 tonnes) of the level of NO_x — a major contributor to smog in cities.

One of the fastest growing sources of CO_2 emissions is the aeroplane [9], with a forecast growth rate of 2–3% annually to 2010, even accounting for significant improvements in fuel efficiency. A study in 1994 by the Worldwide Fund for Nature suggested that world-wide emissions will treble by the year 2040 and the international air transport association (IATA) expects passenger numbers in Europe to treble by 2010. The energy efficiency of telecommunications is dramatically illustrated when one compares the 1994-5 annual CO_2 emissions from British Airways flights of 13 million tonnes, with the 2 million

tonnes emitted as a result of energy consumed to run the entire UK-based BT network [10].

In terms of the commercial benefits that PNOs may gain from the deployment of services/applications, the project team undertook to make calculations using the successive calculation principle [11]. This is a practical decision-making support methodology that covers situations where there is great uncertainty over the data to be input. The basis of the approach is an uncertainty analysis — at each stage of the process, the model requires that the focus is placed on precisely those aspects where there is the greatest level of uncertainty. Indeed, this is the first of the four corner stones of the analysis method. The other corner stones are the evaluation technique itself, a top-down approach, and the identification of all possible aspects of the problem in the general conditions.

The calculation was performed on the basis of a forecast of the opportunities in Western Europe (to include the former East Germany but no other countries from the former Eastern bloc) over the period 1996 to 2010, with 2006 being taken as the representative year. The European Currency Unit (ECU) was chosen as the unit for the calculation, with the analysis focusing on the PNOs' primary activities, inclusive of value-added tax and other fees and related capital costs but before corporate tax, dividends, etc.

For the analysis itself, revenue for the PNOs was considered as the sum of two distinct contributions — that related to sustainable development and that which was not. The former is obtained as a result of an understanding attitude towards sustainable development, while the latter is revenue from all other causes.

In order to simplify the calculation four possible scenarios were defined to take account of two possible developmental levels in the PNO and society. This is represented in Table 6.

Table 6 Scenarios.

	PNOs conservative	PNOs progressive
Conservative Society	TURTLE	DOLPHIN
Progressive Society	POLAR BEAR	CHAMELEON

The naming of the scenarios was more than symbolic. Each animal was chosen for a particular reason:

- TURTLE — lives by habit, incapable of changing — PNOs and society conservative,
- DOLPHIN — intelligent and fast but lives in a relatively stable environment — PNOs progressive,

- POLAR BEAR — subsists in harsh surroundings — society continually changing,
- CHAMELEON — changes its colour to match surroundings — both PNOs and society progressive.

The projected revenue for PNOs generated as a result of an understanding of sustainable development was calculated using the uncertainty principle and the values obtained scaled as shown in Table 7.

Table 7 Scaled revenues from the scenarios.

TURTLE	DOLPHIN
1	1.54
POLAR BEAR	CHAMELEON
4.5	6.5

In other words, assuming that today's position relates to the Turtle scenario, the potential additional commercial benefits in moving towards the Chameleon situation is increased 6.5 times.

In terms of the five opportunity areas that had already been defined, the projected change as one proceeds through the four scenarios is as shown in Table 8.

Table 8 Market share for opportunity areas.

Scenario	Opportunity areas				
	Letter post	Travel options	Info transfer	Consumer Services	Telematics
TURTLE	16%	14%	19%	15%	36%
DOLPHIN	19%	15%	19%	16%	31%
POLAR BEAR	15%	29%	19%	19%	18%
CHAMELEON	13%	30%	21%	19%	17%

As well as forecasting a tremendous potential increase in revenue as one moves from today's situation (Turtle) to one where both the PNOs and society are progressive (Chameleon), the relative levels in each of the identified opportunity areas change. Even where a relative decrease in market share is shown, the actual revenue, however, still goes up because the size of the market has increased so much.

6. Conclusions

The project has demonstrated that there are both challenges and advantages to be gained from the achievement of sustainable development. Since the majority of the world's nations have set sustainable development as a major goal, this means that telecommunications providers can play a significant part in its achievement. In order to do so, however, they will need to take account of their own activities — perhaps to a much greater extent than has taken place previously?

The project has produced not only an environmental charter against which to report progress with its activities, but also a range of standards and measures by which current status can be measured and improvements made. Not the least of the project's achievements, however, has been the forecasting of commercial and societal benefits that are possible through the increased use of telecommunications services. A model of services in opportunity areas has been introduced as an aid towards successful marketing of those services. BT is already in a strong position not only to implement the results of the project but also to take the issues raised even further, and Systems Integration department will be in the forefront of the development and deployment of environmental tools within BT and Concert.

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